

Abdelhamid Attaby, Ph.D.

Staff Software Engineer · Distributed Systems · Technical Leadership

New Cairo, Egypt (remote, any timezone)

[linkedin.com/in/abdelhamidattaby](https://www.linkedin.com/in/abdelhamidattaby) · github.com/abdelhamid-attaby · Google Scholar: 124 citations

SUMMARY

Staff-level software engineer with 15+ years building, shipping, and operating medium-to-large-scale applications, including two years at GitHub (Microsoft) contributing to one of the world's largest monolithic codebases across three engineering teams. Since 2023, has channeled that experience into startups and applied AI — co-founding an early-stage venture and serving as a founding engineer on a multi-agent AI application platform — owning production agentic-AI products hands-on, from architecture through delivery. Full-stack strength across Node.js, TypeScript, Ruby on Rails, React, Go, and Elixir, with deep experience in microservices, distributed systems, API design, and CI/CD, and a daily AI-assisted engineering practice (Claude, Claude Code, LLM APIs). Twice CTO and twice tech lead, with a track record of setting technical direction across teams, architecting platforms from the ground up, and mentoring engineers from early-career to senior — the cross-team ownership and system-wide judgment a Staff role calls for. Holds M.Sc. and Ph.D. degrees in Computer Science and Engineering and is pursuing an MBA (AASST), with research spanning software engineering, computer networks and security, and AI.

CORE STRENGTHS

- Distributed systems & APIs at extreme scale — built the GraphQL APIs behind GitHub Projects; ran performance work and data migrations across hundreds of millions of records.
- Full-stack product engineering — a decade of production Node.js/TypeScript, React, Ruby on Rails, Go, and Python across fintech, healthcare, and developer tools.
- Cross-team technical leadership — CTO twice, tech lead twice; set architecture direction, hired, mentored, and ran engineering teams end-to-end, from design through delivery.
- AI & agent engineering — builds production agent workflows with Claude daily (autonomous research agents, scheduled tasks, tool integrations) and ships with LLM-assisted development.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Go, Ruby, Elixir, PHP, C#

Backend: Node.js, Express, NestJS, Phoenix (Elixir), Ruby on Rails, Django, Flask, Laravel, REST APIs, GraphQL, Socket.IO, Stripe, Twilio

Frontend: React, Next.js, React Native, Android, Redux, MobX, HTML5, CSS3

Databases: PostgreSQL, MongoDB, MySQL, Microsoft SQL, Redis, Firebase, Google Firestore

Cloud & DevOps: AWS, Google Cloud Platform, Azure, Docker, Kubernetes, Terraform, serverless (AWS Lambda, Google Cloud Functions), Nginx, Cloudflare, DigitalOcean, Netlify, CI/CD (GitHub Actions, CircleCI, Jenkins), Linux, Datadog, Sentry

AI / LLM: Claude, Claude Code, Claude API, ChatGPT, GitHub Copilot, Cursor AI, LLM APIs (Anthropic, OpenAI), AI agents & agentic AI, AI agent orchestration, agentic coding, AI integration, LLM-assisted engineering

Practices: Full-stack, microservices, distributed & event-driven systems (Apache Kafka, RabbitMQ, NATS, Google Pub/Sub, Protobuf), SaaS, software architecture, DevOps, automated testing (Jest), web scraping, Agile/Scrum, technical leadership & mentoring

PROFESSIONAL EXPERIENCE

Founding Engineer — Flowdia (flowdia.ai)

Dec 2025 – Present

Remote · AI Application Platform

- Founding engineer on an AI application platform where a team of specialized AI agents plans, designs, builds, and ships production web apps end-to-end — each with a real backend, dedicated PostgreSQL database, and APIs.
- Building the core product hands-on across an Elixir and TypeScript/Node.js stack: agent orchestration, LLM integration, and the services behind the multi-agent build-and-ship pipeline.

- Engineering the system that turns natural-language product intent into deployed, production-ready applications with persistent project memory across sessions.

Co-founder — Bardy

May 2023 – Dec 2025

Remote · SaaS

- Co-founded an early-stage SaaS startup building a real-time visual feedback and collaboration platform — comment directly on live websites, videos, PDFs, and images, edit HTML/CSS in place, and auto-capture screenshots, delivered via web app and a Chrome extension.
- Built the product end-to-end as one of two co-founders: Elixir backend services and proxy, JavaScript/React frontend and Chrome extension, a TypeScript screenshot service on AWS Lambda, all provisioned with Terraform.
- Owned product and engineering direction across a multi-tenant SaaS; made the disciplined decision to wind the company down in late 2025, carrying the learnings directly into AI product engineering.

Senior Software Engineer — GitHub (Microsoft)

Apr 2021 – Apr 2023

Remote

- Built and shipped GitHub Projects — the planning-and-tracking experience used by organizations and developers worldwide — as part of the Memex team.
- Contributed across the github/github monolith, one of the world's largest codebases, working with three engineering teams (Projects, Special Projects/Runtime, internal engineering documentation platform).
- Delivered saved views, user-owned projects, the project GraphQL APIs, fast issues, the iterations field, and the issue sidebar.
- Worked on large-scale performance, distributed data migrations, and platform reliability using Ruby on Rails, Go, React, TypeScript, and MobX.

VP of Engineering — Wuilt

Sep 2021 – Sep 2022

Remote · SaaS / Website Builder

- Set engineering strategy and technical direction at a regional SaaS/website-building company, partnering with leadership on roadmap and execution.
- Architected core platforms from the ground up and established engineering standards, delivery processes, and CI/CD.
- Built and scaled the engineering team through hiring, mentoring, and code review.

Chief Technology Officer — Aster

Aug 2020 – Sep 2021

Remote · Fintech

- Led the engineering team to deliver product requirements; set technical strategy and direction with the leadership team; owned the team budget and tooling.
- Architected the backend on a microservices model and built APIs aggregating users' wealth into dashboards and charts.
- Integrated numerous third-party financial APIs to extend platform capability.

Senior Consultant — Andela

Jan 2020 – Aug 2020

Remote

- Senior engineer for Simple Habit, Inc. (San Francisco), shipping a high-quality meditation product through engineering best practices.
- Designed and conducted technical interviews for senior and principal candidates; mentored mid-level and senior engineers.

Technical Team Lead — Pocketmade LLC (via Toptal)

Feb 2019 – Apr 2020

Remote · Healthcare logistics

- Led frontend and backend teams building and deploying the Grow Healthy platform, integrating multiple systems and APIs.
- Integrated Twilio, OnFleet, Zendesk, and Google Maps; built order processing on Google Firestore, Scheduler, Pub/Sub, and Cloud Functions.
- Authored unit and end-to-end tests with Jest; automated deployments with GitHub Actions.

Co-founder & Chief Technology Officer — Coligo Technologies

Dec 2016 – Dec 2019

Cairo, Egypt · EdTech

- Co-founded an EdTech company and designed the core architecture of its learning management system.
- Directed cross-platform engineering across web, Android, and iOS through an Agile process.
- Built fully automated CI/CD pipelines on CircleCI for containerized applications using Docker, Netlify, and AWS.

Software Technical Lead — So Visions

Mar 2014 – Nov 2016

Cairo, Egypt

- Led a team building enterprise Odoo (OpenERP) modules and a RESTful API server integrating Odoo with third-party applications.
- Established team-wide engineering workflows and CI/CD pipelines on CircleCI using Docker and DigitalOcean.

Earlier Experience — Toptal · So Visions · Arabia Inform

2011 – 2019

- Node.js Back-end Engineer (BeatFlyer, via Toptal, 2019): built RESTful APIs for a fleet-management platform with multiple authentication schemes and Jest coverage.
- Software Engineer (So Visions, 2012–2014): developed accounting, purchasing, inventory, and sales modules of a full ERP system.
- Software Engineer (Arabia Inform, 2011–2012): built backend crawlers using the Twitter Search and YouTube Data APIs.

EDUCATION

MBA in Business Administration

In progress, expected 2027

Arab Academy for Science, Technology & Maritime Transport (AAST), Cairo

Ph.D. in Computer Science and Engineering

2016 – 2019

Egypt-Japan University of Science and Technology (E-JUST)

M.Sc. in Computer Engineering

2010 – 2015

Benha University

B.Sc. in Computer Engineering (Very Good; ranked 2nd in class)

2004 – 2009

Faculty of Engineering at Shoubra, Benha University

SELECTED PUBLICATIONS

Peer-reviewed work in wireless network testbeds and applied data security (124 citations, Google Scholar):

1. WiPi: A Low-Cost Large-Scale Remotely-Accessible Network Testbed. IEEE Access, 7 (2019).
2. WiPi: A Low-Cost Heterogeneous Wireless Testbed for Next Generation Applications. IEEE VTC Spring (2018).
3. Data Hiding Inside JPEG Images with High Resistance to Steganalysis (DCT-M3). Ain Shams Engineering Journal, 9(4) (2018).

Research interests: Software engineering and large-scale distributed systems; computer networks and security; artificial intelligence (LLM-based and agentic systems)

CERTIFICATIONS & LANGUAGES

Certifications: Cisco Certified Network Professional (CCNP); CCNA; Moving from Developer to Engineering Manager; Managing Technical Professionals

Languages: JavaScript, TypeScript, Python, Go, Ruby, Elixir, PHP, C#